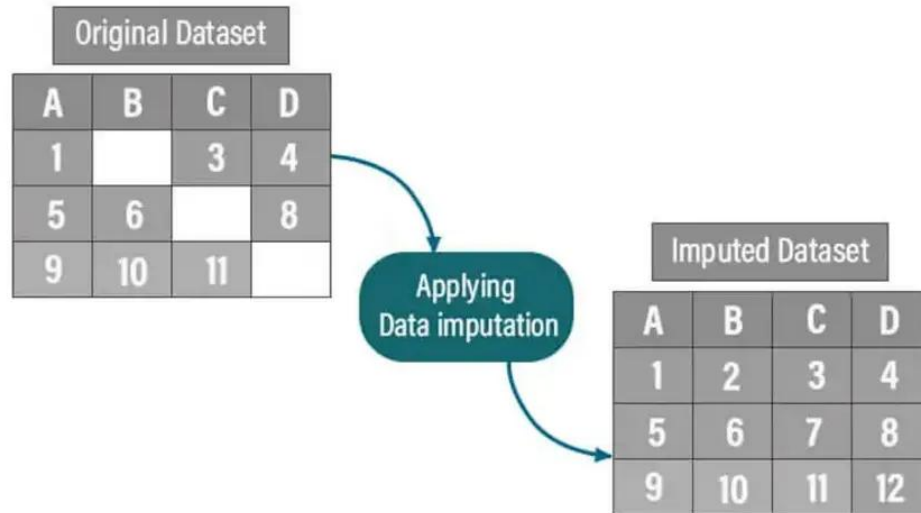


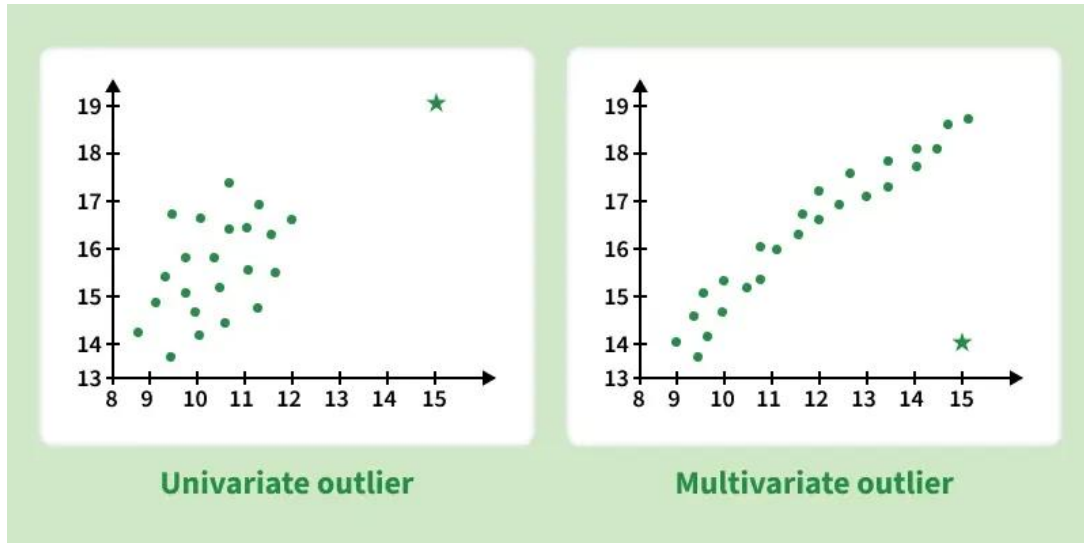
Handling the Void (Imputation Strategy)

Data Imputation



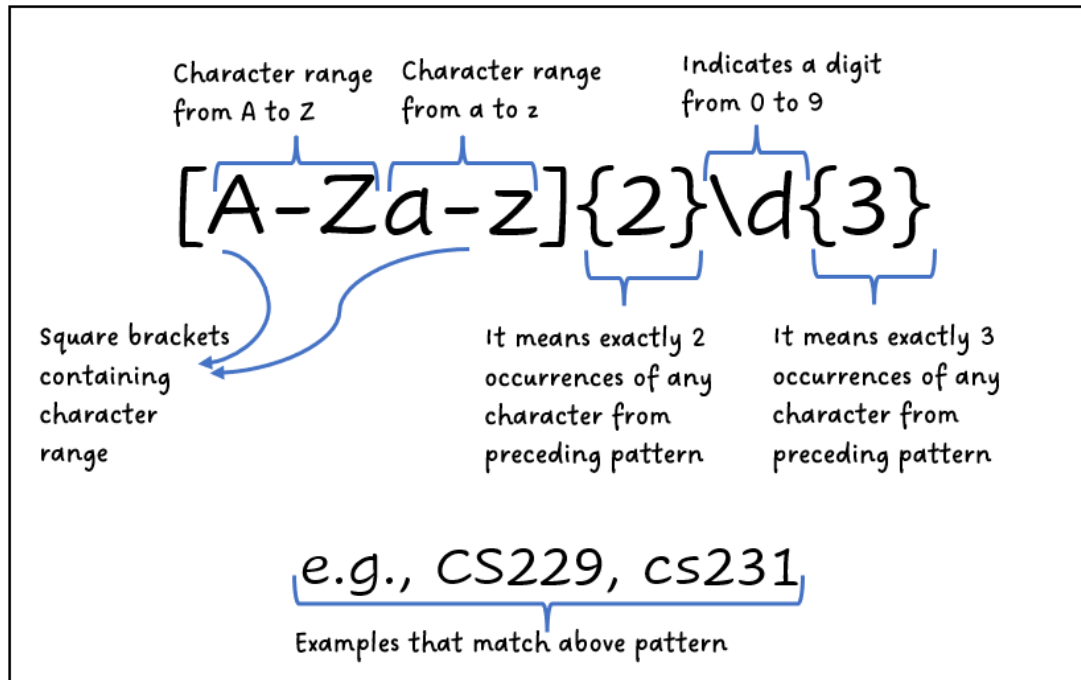
- **The Problem:** Real-world data is rarely complete. Deleting every row with a missing value can destroy 50% of your dataset.
- **The Solution: Scientific Imputation.**
 - **Mean:** Best for normally distributed data.
 - **Median:** Best for skewed data (like salaries) as it ignores outliers.
 - **Mode:** The "Most Frequent" choice for categorical data (like City names).

Outlier Engineering (Finding the "Impossible")



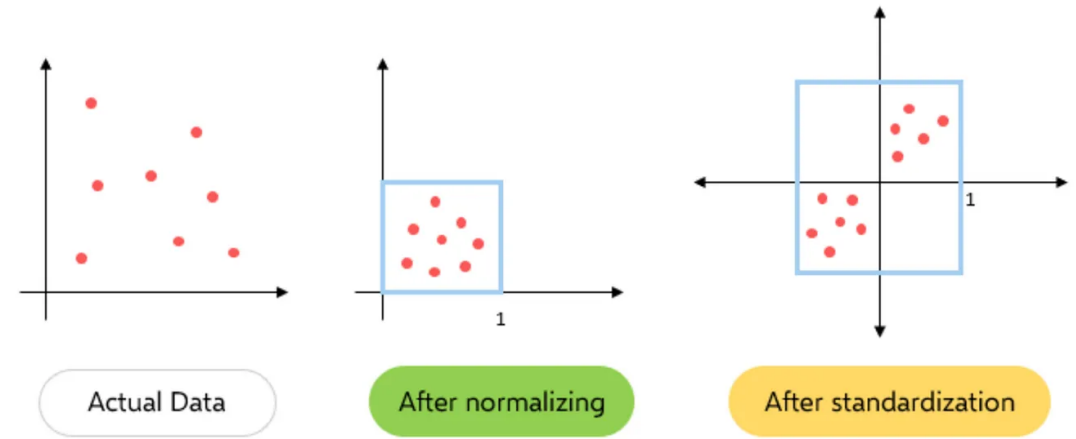
- **What are Outliers?:** Data points that don't fit the pattern (e.g., an "Age" of 250 or a "Price" of -10).
- **Detection Methods:**
 - **Z-Score:** Measures how many standard deviations a point is from the mean.
 - **IQR (Interquartile Range):** The middle 50% of the data. Anything significantly outside the "whiskers" of a boxplot is flagged.
- **Outcome:** We don't always delete outliers; sometimes we "clip" them to a maximum logical value.

String Cleaning (The Power of Regex)



- **Messy Text:** Computers see "Peshawar", "peshawar", and "Peshawar " as three different cities.
- **String Methods:** Using `.str.strip()` and `.str.lower()` to unify text.
- **Regular Expressions (Regex):** Powerful pattern-matching logic to extract phone numbers from text or fix email typos automatically.

Standardization & Scaling



- **The Magnitude Problem:** A computer thinks a "Salary" of 100,000 is "more important" than an "Age" of 30 just because the number is bigger.
- **Standardization:** Centering data around 0 with a standard deviation of 1.
- **Normalization:** Squishing all values between 0 and 1.
- **Goal:** Ensuring the model treats all "features" (columns) with equal mathematical respect.